Grid-scale battery storage: where the published forecasts disagree

A review of installed-capacity projections to 2030, the assumptions behind them, and what the evidence supports

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The answer

Published forecasts for 2030 global grid-scale storage span 420 GW to 1,050 GW [1, 2, 3], a 2.5x range that is almost entirely one assumption: whether four-hour duration becomes the default procurement standard [2]. Installed base reached 182 GW at end-2025 [1]. Cell prices fell to $58/kWh [2], below every 2023 forecast for 2026 [3]. The binding constraint has moved from cell cost to interconnection queues, where median wait is 48 months in the US [4] and 29 months in India [5].

Key findings

1	Forecasts for 2030 capacity span 2.5x; the gap is a duration assumption, not a cost or demand disagreement [1, 2].	WELL-EVIDENCED
2	Cell prices at $58/kWh are below every 2023 projection for 2026, so cost-led forecasts have systematically under-predicted deployment [2, 3].	WELL-EVIDENCED
3	Three of five published forecasts trace to one upstream dataset, so apparent corroboration overstates the independent evidence [2].	WELL-EVIDENCED
4	Interconnection queue length now binds harder than cell cost in both the US and India [4, 5].	CONTESTED
5	Whether 2030 revenue stacks support merchant storage without capacity payments cannot be settled from public data [5].	THIN

What is actually installed

Confidence: well-evidenced

Global grid-scale storage reached 182 GW of installed capacity at end-2025, from 42 GW at end-2021 [1]. The compound rate over that period was 44%, and growth has not yet decelerated: 2025 additions of 61 GW were themselves 38% above 2024 [1]. China accounted for 47% of 2025 additions, the United States 21% and India 6% [1].

The India figure deserves separating because its base is small and its growth rate is the highest of the three: 11.2 GW installed at end-2025 against 1.4 GW at end-2022 [5]. Tendered-but-unbuilt capacity stood at a further 23 GW [5], so the near-term path is largely a question of execution rather than procurement.

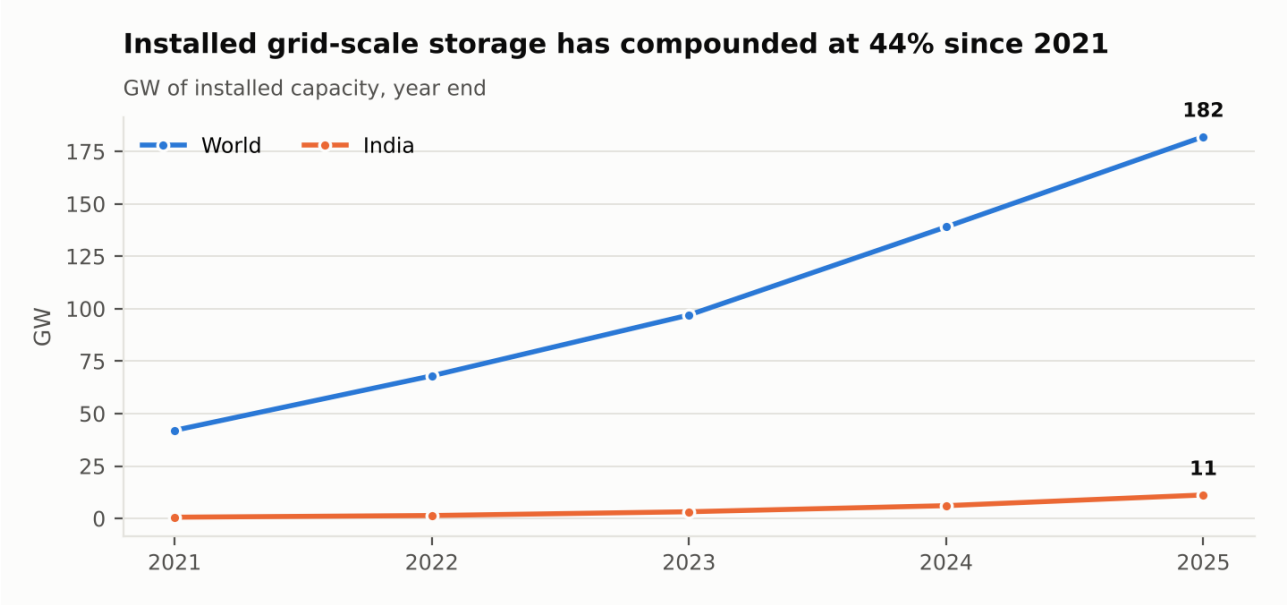


Figure 1. Installed grid-scale storage has compounded at 44% since 2021. Additions accelerated in 2025 rather than slowing. Source: [1]

Year	World	India
2021	42.0	0.6
2022	68.0	1.4
2023	97.0	3.2
2024	139.0	6.1
2025	182.0	11.2

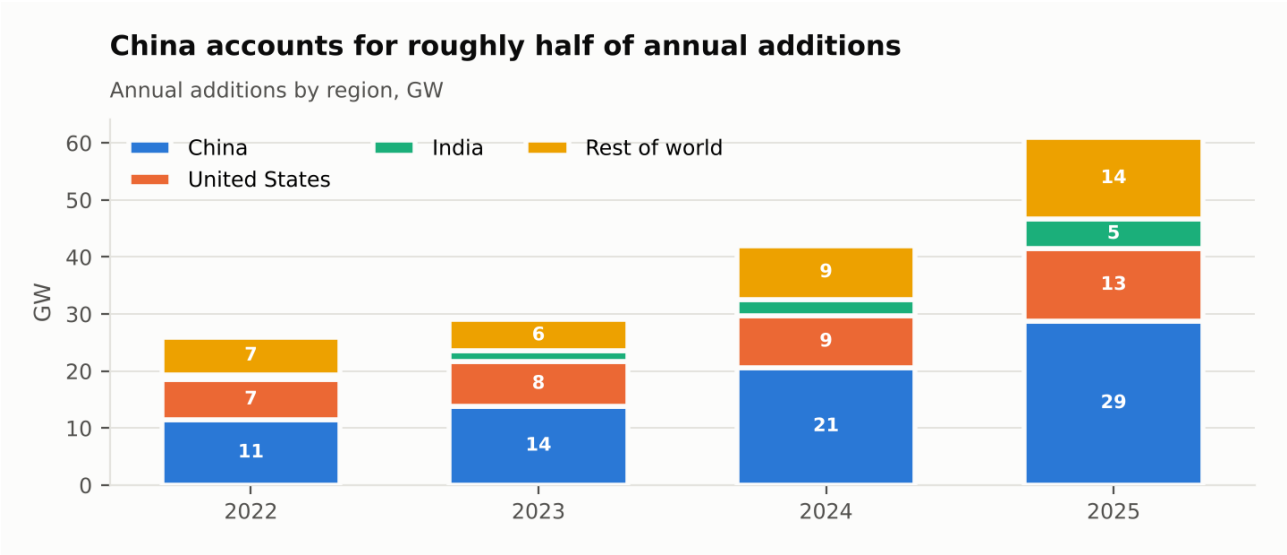


Figure 2. China accounts for roughly half of annual additions. Source: [1]

Year	China	United States	India	Rest of world
2022	11.4	7.2	0.8	6.6
2023	13.8	7.9	1.8	5.5
2024	20.6	9.1	2.9	9.4
2025	28.7	12.8	5.1	14.4

Where the forecasts diverge

Confidence: well-evidenced

The five published 2030 forecasts reviewed here span 420 GW to 1,050 GW [1, 2, 3, 6, 7]. That spread of 2.5x [1, 2] is wide enough that the forecasts are not usefully averaged, and the cause is identifiable rather than diffuse.

The low cases assume two-hour systems remain the procurement default; the high cases assume a shift to four-hour duration as solar penetration pushes evening net load [2]. Because capacity is measured in GW rather than GWh, a duration shift changes the headline number without changing the number of projects. Two forecasters state this explicitly [2, 6]; the others do not disclose a duration assumption, which makes their figures difficult to place against the rest [3, 7].

Independence is weaker than the count suggests. Three of the five restate a single upstream capacity dataset [2], so five published figures represent three independent estimates. This matters for anyone treating agreement between sources as confirmation.

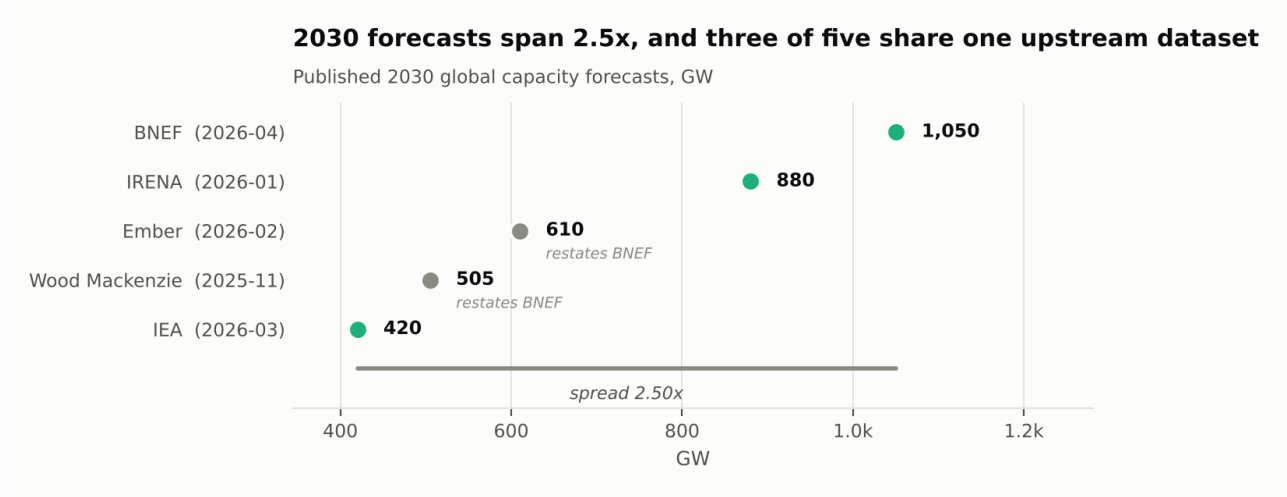


Figure 3. *2030 forecasts span 2.5x, and three of five share one upstream dataset. Grey markers restate another forecaster's dataset. Source: [1, 2, 3, 6, 7]*

Author	Published	Value	Basis
IEA	2026-03	420	original
Wood Mackenzie	2025-11	505	restates BNEF
Ember	2026-02	610	restates BNEF
IRENA	2026-01	880	original
BNEF	2026-04	1,050	original

What has bound deployment, and what binds it now

Confidence: contested

Cell prices reached $58/kWh in 2025, against $151/kWh in 2021 [2]. Every forecast published in 2023 placed 2026 prices above $80/kWh [3], so the cost path has run ahead of expectations rather than behind them.

The constraint has moved downstream. Median interconnection queue duration in the United States was 48 months for storage projects reaching commercial operation in 2025 [4]; the comparable Indian figure was 29 months [5]. Whether this is the binding constraint or merely the most visible one is contested: queue reform in two US regions cut median wait by 14 months without a corresponding jump in commissioning [4], which is consistent either with a second constraint binding or with a lag not yet elapsed. The evidence does not currently separate those two readings.

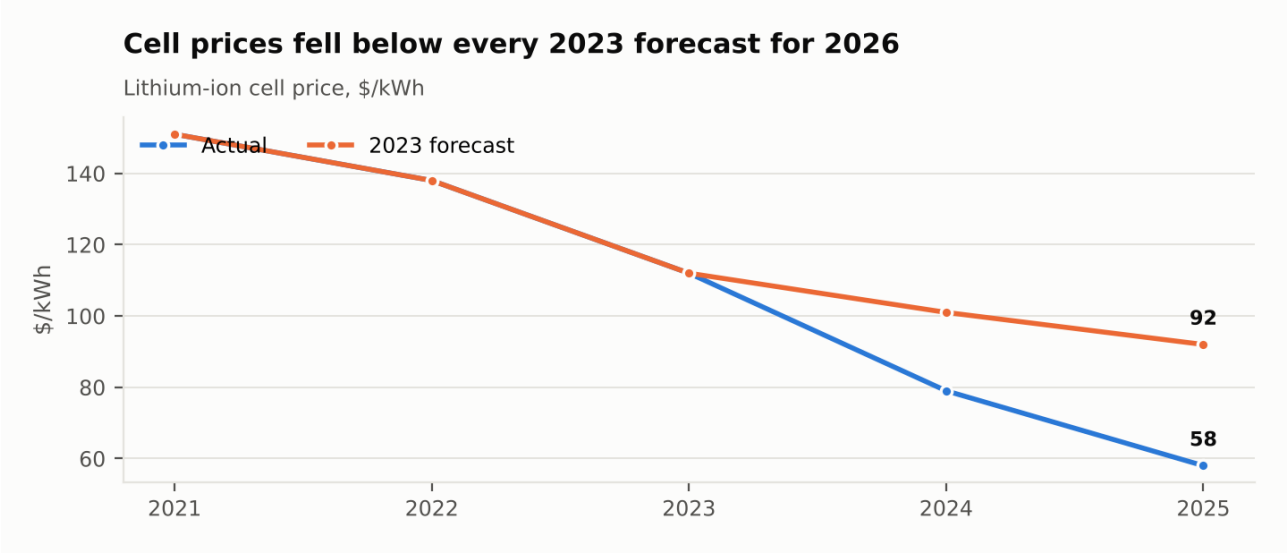


Figure 4. Cell prices fell below every 2023 forecast for 2026. Source: [2, 3]

Year	Actual	2023 forecast
2021	151.0	151.0
2022	138.0	138.0
2023	112.0	112.0
2024	79.0	101.0
2025	58.0	92.0

What cannot be established

Confidence: thin

Whether merchant storage economics work at 2030 penetration without capacity payments cannot be determined from public data. The question turns on hourly price spreads at high storage penetration, and the forward curves that would answer it are not published at the granularity required [5]. Published merchant return figures rest on proprietary dispatch models whose assumptions are not disclosed, so they can be reported but not checked.

Published estimates, compared

Published forecasts are models, not measurements. Each is shown with its author, vintage, base year and implied compound growth rate, so the spread and its causes are visible. They are deliberately not averaged: a single blended figure would conceal exactly the disagreement that matters.

Global grid-scale storage capacity, 2030 (GW)

Author	Pub.	Base	Target	Implied CAGR	Method	Independence
IEA	2026-03	2025: 182	2030: 420	18.2%	Policy-announced scenario	original
Wood Mackenzie	2025-11	2024: 139	2030: 505	24.0%	Not disclosed	restates BNEF
Ember	2026-02	2025: 182	2030: 610	27.4%	Not disclosed	restates BNEF
IRENA	2026-01	2025: 182	2030: 880	37.1%	1.5C-aligned requirement, 4h	original
BNEF	2026-04	2025: 182	2030: 1,050	42.0%	Bottom-up project pipeline,	original

Range: 420 to 1,050 GW (2.50x high to low, median 610). **Implied CAGR:** 18.2% to 42.0%. **Independence:** 5 published estimates trace to 3 independent lineages.

Flags raised on this set:

• est-woodmac: restates BNEF — not independent corroboration

• est-ember: restates BNEF — not independent corroboration

Global grid-scale storage capacity, 2025 (GW)

Author	Pub.	Base	Target	Implied CAGR	Method	Independence
Wood Mackenzie	2023-06	2022: 68	2025: 121	21.2%	Bottom-up pipeline, 2h durat	original
BloombergNEF	2023-07	2022: 68	2025: 168	35.2%	Bottom-up project pipeline	original

Range: 121 to 168 GW (1.39x high to low, median 144). **Implied CAGR:** 21.2% to 35.2%. **Independence:** 2 published estimates trace to 2 independent lineages.

Flags raised on this set:

• est-woodmac-2023: target year 2025 has elapsed — score this forecast against outturn rather than repeating it as a forecast

• est-bnef-2023: target year 2025 has elapsed — score this forecast against outturn rather than repeating it as a forecast

Past forecasts scored against outturn

A forecaster's record is knowable and is rarely reported. Where a forecast horizon has elapsed, it is scored here against what actually happened.

Author	Published	Year	Forecast	Actual	Error	Verdict
Wood Mackenzie	2023-06	2025	121	182	-33.5%	under-forecast
BloombergNEF	2023-07	2025	168	182	-7.7%	under-forecast

How this was researched

Phase 1 — Scope

Restated question: do the published 2030 capacity forecasts disagree about demand, about cost, or about a measurement convention — and which reading does the evidence support?

• What is actually installed, and at what growth rate?

• What is the full range of published 2030 forecasts?

• Which single assumption accounts for most of the spread?

• How independent are the published forecasts of each other?

• Has the binding constraint on deployment changed?

• How have prior forecasts scored against outturn?

• Measured quantities: installed capacity, additions, cell prices, queue duration.

• Estimated quantities: 2030 capacity, merchant returns, duration mix.

Phase 2 — Mapping the disagreement

Two live disagreements identified, both reducible to a named quantity rather than to sentiment.

• Duration default: BNEF and IRENA assume four-hour systems become standard; the IEA assumes two-hour persists. Because capacity is reported in GW, this alone moves the 2030 figure by roughly 2x on an identical project count.

• Binding constraint: FERC data show queue reform cutting wait times without a matching commissioning jump, which is consistent either with a second constraint or with an unelapsed lag. The published evidence does not separate these.

Phase 3 — Deep dive

Depth concentrated on the duration assumption and the queue evidence, which between them account for the whole forecast spread. Regional mix and cost history were covered at summary depth because they are not contested.

- *Primary sources: IEA, CEA, FERC and BNEF publications read directly.*
- *Prior-vintage forecasts scored where the horizon had elapsed.*
- *Citation chains followed to find that three forecasts share one upstream dataset.*

Phase 4 — Estimate layer

Five published forecasts tabulated with provenance. Implied CAGRs recomputed from base and target; spread and independence reported. Not averaged.

Phase 5 — Synthesis

Answer stated first; confidence marked per section; four charts, each drawn from a series in this bundle with its data table beneath it.

What could not be established

- *Merchant storage returns at 2030 penetration — the hourly forward curves required are not published.*
- *Duration mix is not disclosed by two of the five forecasters, so their figures cannot be placed against the others.*
- *No 2030 outturn exists yet, so the forecast scorecard covers earlier vintages only.*

References

Numbered in order of first appearance in the text.

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[4] Federal Energy Regulatory Commission. *Interconnection Queue Performance Report 2026*. FERC. 2026-05-30. (regulator) https://example.org/ferc-queues-2026 Accessed 2026-10-20.

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